

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester I Examination, 2022-23

Paper No. STB-101: Statistical Methods and Probability

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer **five** questions including Question 1 which is compulsory. Symbols have their usual meanings.*

Question 1.

- (a) What do you mean by measures of central tendency? Also, describe the procedures of construction of a frequency curve, frequency polygon, and histogram.
 - (b) Define geometric mean. Also, write down its merits and demerits. Further, let G_1 and G_2 be the geometric means of two individual series and n_1 and n_2 be their sizes respectively, then derive the formula for the geometric mean of the combined series.
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Question 2.

- (a) Define moments about origin and moments about arithmetic mean. Further, derive the relation between moments about mean in terms of moments about origin.
 - (b) Define mean square deviation and variance. Also, calculate the mean deviation from the mean and standard deviation of the series $a, a + d, a + 2d, \dots, a + 2nd$ and verify that the latter is greater than the former.
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Question 3.

- (a) Give the axiomatic definition of probability. Next, if A and B be any two events of a random experiment, then find the probability of happening of at least A or B .
 - (b) If a and b be any two non-negative quantities such that $a + b = 2n$, then find the probability that their product is not less than $\frac{3}{4}$ times their greatest product.
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Question 4.

- (a) State and prove a necessary and sufficient condition for any two events A and B of a random experiment such that $P(A) > 0$ and $P(B) > 0$ to be independent events.
 - (b) Define conditional probability. Further, if a certain drug manufactured by a company is tested chemically for its toxic nature. Let the event “the drug is toxic” be denoted by E and the event “the chemical test reveals that the drug is toxic” be denoted by F . Let $P(E) = \lambda$, $P(F|E) = 1 - \lambda$, and $P(F^c|E^c) = 1 - \lambda$. Then prove that the probability that
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the drug is not toxic given that the chemical test reveals that it is toxic, is independent of λ .

Question 5.

- (a) If $E_1, E_2, \dots, E_n$ be any n mutually exclusive events of a random experiment such that $P(E_i) \neq 0$ for all $i = 1, 2, \dots, n$, and let A be any arbitrary event which is a subset of $\bigcup_{i=1}^n E_i$, then calculate $P(E_i|A)$ for all $i = 1, 2, \dots, n$ in the case when $P(A) > 0$.
- (b) There are two bags A and B. A contains n white and 2 black balls, and B contains 2 white and n black balls. One of the two bags is selected at random and 2 balls are drawn from it without replacement. If both the drawn balls are white and the probability that the bag A was used to draw the balls is $\frac{6}{7}$, then find the value of n .
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Question 6.

- (a) Define mathematical expectation of a random variable. Further, state and prove the multiplication theorem of expectation for any two independent random variables X and Y .
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Question 7.

- (a) Define probability density function (p.d.f.) of a continuous random variable. Next, if X denotes the diameter of an electric cable and has the following probability density function:

$$f(x) = 6x(1 - x), \quad 0 < x < 1$$

and 0 otherwise. Then:

- (i) check whether $f(x)$ is a p.d.f., and
- (ii) determine a number b such that $P(X < b) = P(X > b)$.
- (b) If the joint p.d.f. of X and Y is:

$$f(x, y) = \frac{8}{9}xy, \quad 1 \leq x \leq y \leq 2$$

and 0 otherwise, then find the conditional p.d.f. of Y given $X = x$ and that of X given $Y = y$.

- (c) Define joint p.m.f. and joint p.d.f. Next, the joint probability distribution of two random variables X and Y is given by:

$$P(X = 0, Y = 1) = \frac{1}{4}, \quad P(X = 1, Y = -1) = \frac{1}{2}, \quad P(X = 1, Y = 1) = \frac{1}{4}$$

Then find:

- (i) the marginal probability distributions of X and Y , and
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- (ii) the conditional probability distribution of X given $Y = y$.
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Question 8.

- (a) Derive the expression for the variance of the linear combination of any n random variables.
- (b) Calculate the expectation of the number of failures preceding the first success in an infinite series of independent trials with constant probability of success, say p , in each trial.